Student Name:
Student Number:

Albalqa Applied University
Faculty of Engineering Technology

Med Exam, Second Semester: 2016/2017
Mechatronics Department

Course Title: Power Electronics
Marks: 40
Lecturers: Prof. Dr.Qazem M. Jaber & Dr Ghazi .M. Qaryouti

Date: 21/3/2017
Time: 1:15 hour
No. of pages: 2

Question 1: *(14marks)*

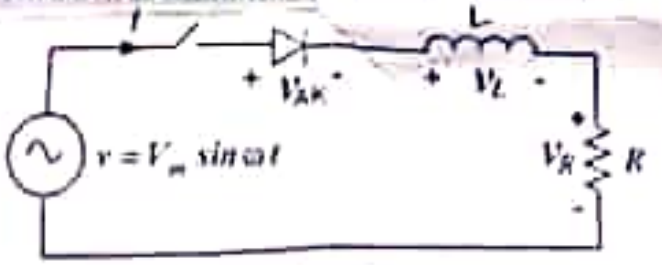
1-Draw the power circuit of three phase controlled rectifier Type Bridge (6 pulses), with ab phase is reference.
2-Draw the line to line waves versus time of power circuit of 6 pulses.
3-Draw the firing signals of thyristores with delay angle (α) = zero.
4-Draw the output instantaneous voltage versus time for circuit of item 1 and item3, with assigning the all necessary number of the thyristor and phase voltage for each interval.
5-Find the anode cathode voltage of thyristot 1 at $\omega t = 30^{\circ}$ and $\omega t = 70^{\circ}$.
6-Scetch the average and the rms voltages the output instantaneous voltage wave versus time.

Question2: *(10marks)*

What is the differences for the following:
1-The lathing and the holding anod currents of thyristor.
2-The temperature of the transformer is fed the half-wave single phase controlled rectifier and the full-wave controlled rectifier with the same resistive load.
3-The power factot of the full -wave single phase controlled rectifier with α = 0°and the full-wave controlled rectifier with α =30° (the load for the two circuits is resistive load).
4-The rms value of the resistive load for the controlird rectifier (3 pulses) and the controlled rectifier (6 pulses)with the rms current of the thyristor is = 3 A.
5-The body of the power thyristor and the the small power thyristor.

Question3: *(8marks)*

Derive the equation of output current that is used to find the value of the extinction angle (β) ,and the average output current (I_o) for the circuit shown in the figure below ,after the switch is closed at t = 0.



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The singl phase full-wave centre tapped controlled rectifier has the secondary voltage of 110V RMS at 50Hz and a 10 Ω load resistor. The delay angle is $\alpha = 30^{\circ}$ Determine:

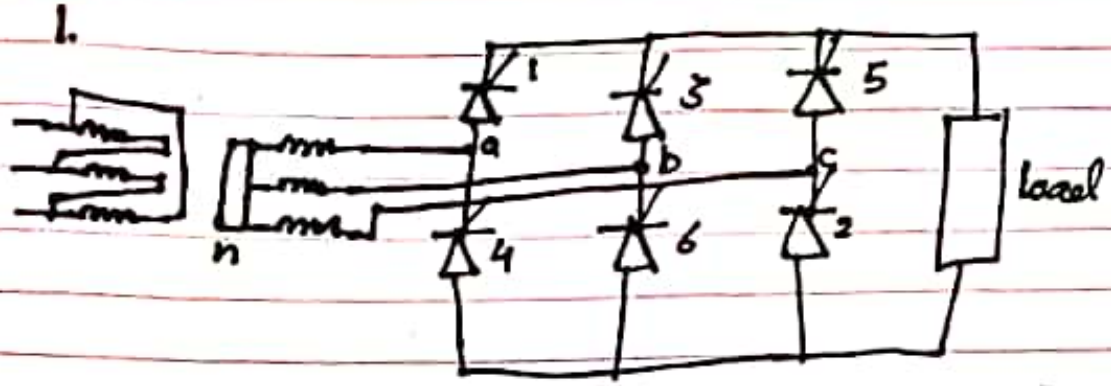
1- The average and RMS load current value.
2- The average and RMS thyristor current value.
3- The efficiency.
4- The Peak Invers Voltage (PIV) of the thyristor.

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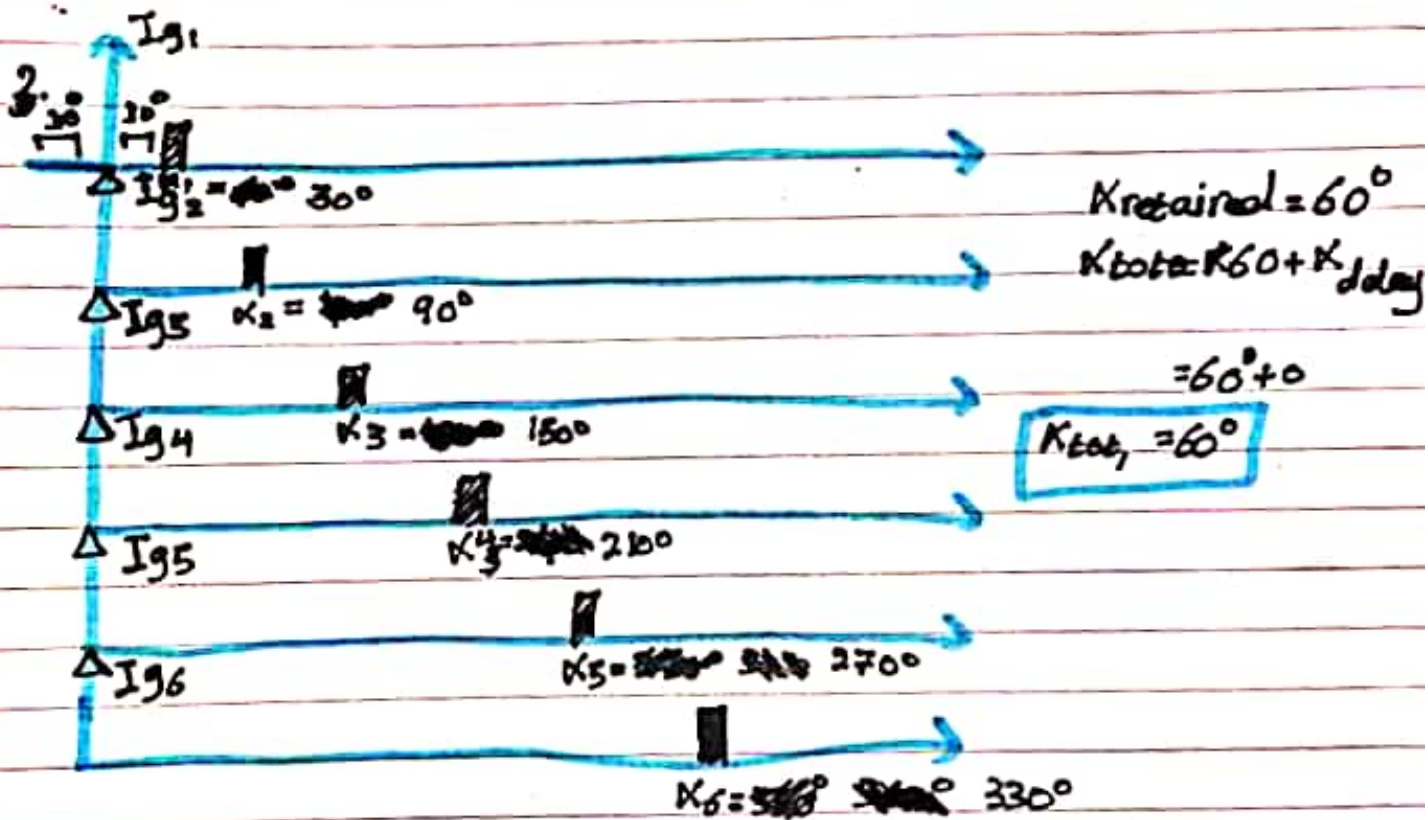
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Question 1

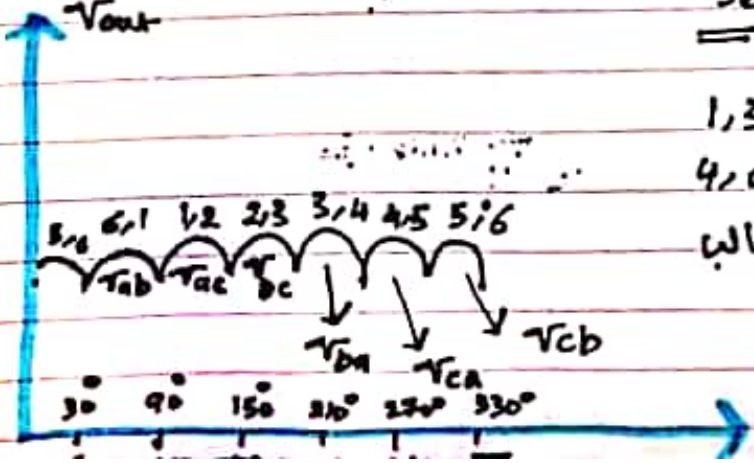
1.



2.



4.



seq ← ارقام الثايرستورات حسب ال

← اسماء ال Voltage ⇦

1,3,5	الثايرستورات الموجبة
4,6,2	" السالبة

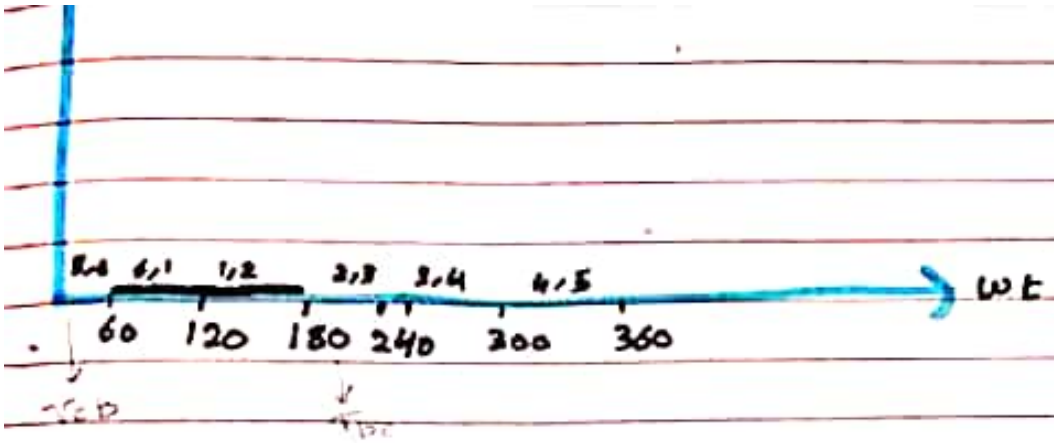
بنحط الموجب بالاول بعدين السالب

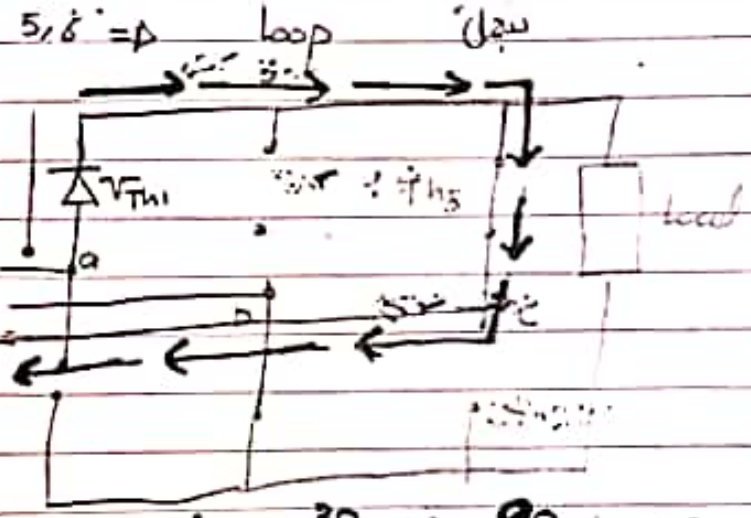
مثلاً ⇦

6,1 ← سالب مستبعد

6,1 ← موجب مقبول

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30 90

$(0 \rightarrow 60) \rightarrow (0+30 \rightarrow 60+30) \rightarrow$

$-V_{an} + V_{TH_1} + V_{cn} = 0$

$V_{TH_1} = V_{an} - V_{cn}$

$V_{TH_1} = V_{ac}$

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$$-V_{TH} + V_{b1} \Rightarrow$$

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$$0 = V_m \sin(B-\phi) - \frac{V_m}{R} \sin\phi \, e^{-\frac{BR}{\omega L}}$$

$\omega t = B$

$$t = \frac{B}{\omega}$$

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$$= \frac{-\sqrt{2}}{2\pi} * 110 \left[-1 - \frac{\sqrt{3}}{2} \right]$$

$$V_{av} = +\frac{\sqrt{2}}{2\pi} 110 \left[1 + \frac{\sqrt{3}}{2} \right]$$

$$I_{av} = \frac{V_{av}}{R}$$

$$I_{av} = \frac{\frac{\sqrt{2}}{2\pi}(110)\left(1+\frac{\sqrt{3}}{2}\right)}{10} \text{ A}$$

$$I_{rms} = \left(\sqrt{\frac{1}{2\pi} \int_{\alpha}^{\pi} V_m^2 \sin^2 \omega t \, d\omega t} \right) / R$$

$$= \left(\sqrt{\frac{V_m^2}{2\pi} \int_{\alpha}^{\pi} \frac{1}{2}(1 - \cos 2\omega t) \, d\omega t} \right) / R$$

I_{rms} Th =

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= [illegible] I

$$\eta = \left(\frac{I_{av\,load}}{I_{rms\,load}}\right)^2$$

تقويض مباشر

تقويض جذر

④ pIV [illegible] = [illegible] | Vm]

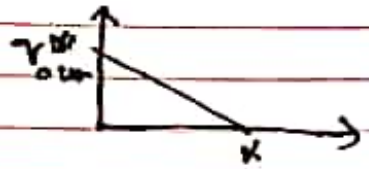
[illegible]

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-Thyristor can be turned off or returned to Forward bloking only if Farward current falls below a low level current called Holding curren - Holding current is less than latching Current	- it is the minimum value of anode current which it must attain during turn on process to maintain conduction when gate pulse is removed. - Latching current is the 2 to 3 times of holding current.

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③ α =D depend on the output voltage:



power factor $\propto$ V_{out}

$\propto \frac{1}{\alpha}$

power factor ای انه کلما زادت قیمة α قل ال

$\alpha = 0$	$\alpha = 30^\circ$
power factor is hig higer	power factor is less.

④

three pulse	6 pulse
$I_{rms\ Load} = \sqrt{I_{1rms}^2 + I_{3rms}^2 + I_{rms5}^2}$	$I_{rms\ Load} = \sqrt{I_{1rms}^2 + I_{2rms}^2 + I_{3rms}^2 + I_{4rms}^2 + I_{5rms}^2 + I_{6rms}^2}$
$= \sqrt{3}\ I_{rms\ th}$	$= \sqrt{6}\ I_{rms\ th}$
$= \sqrt{3} * 3$	$= \sqrt{18} * 3$
$I_{rms\ Load} = 3\sqrt{3}A$	$I_{rms\ Load} = 3\sqrt{6}A$

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مهم جداً

=> Draw V_{L-L} voltages consider that V_{bn} is reFFerence Voltage.

reference: يبساطة معنى كلمة reference يعني انه الـ (phase shift) Zero

V_{bn} فبتعبير V_{bn} هي المرجعية

$V_{bn} = V_m \sin(\omega t - 0)$

+sequence طبعاً احنا بنمشي بالـ +sequence يعني a → b → c

بما انه b هي المرجعية

b → c → a

phase Voltages:

$V_{bn} = V_m \sin \omega t$

$V_{cn} = V_m \sin(\omega t - 120)$

$V_{an} = V_m \sin(\omega t + 120) = -240°$

طالب الـ V_{L-L} ⇐ ممكن

$V_{ab} = V_{an} - V_{bn}$

$= V_m \sin(\omega t + 120) - V_m \sin(\omega t - 0)$

$= V_m \angle 120° - V_m \angle 0°$

$= V_m (\angle 120° - \angle 0°)$

$= V_{bn} \angle 120° - \angle 0°) \Rightarrow$ على الآلة الحاسبة

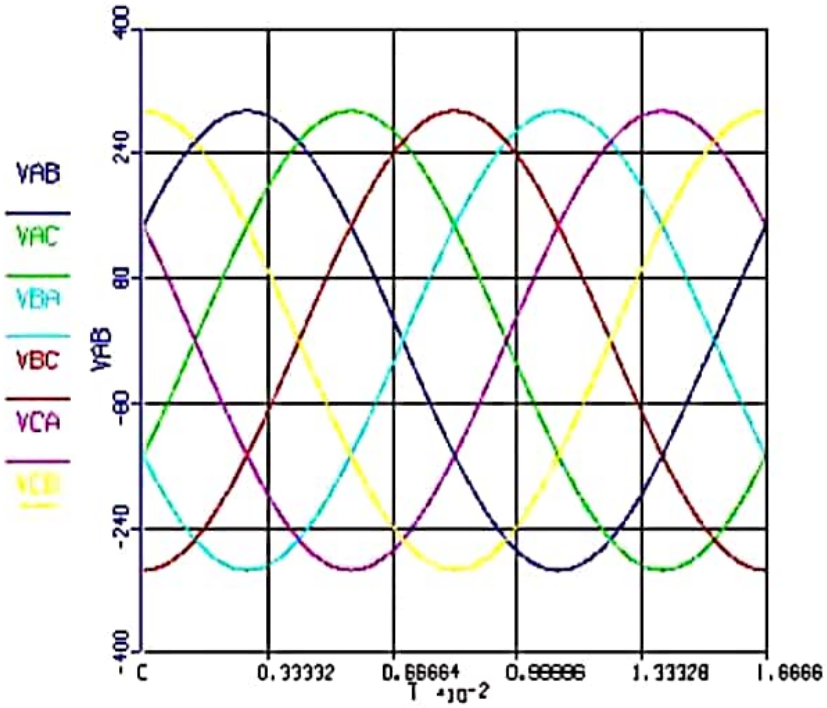
$= V_{bn} (\sqrt{3} \angle +150°)$ ← هي نفسها V_{bn}

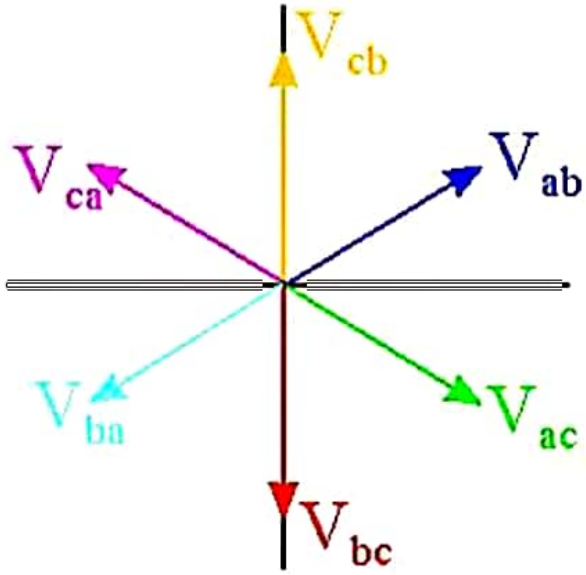
$\boxed{V_{ab} = \sqrt{3} V_{bn} (\sin(\omega t + 150°))}$ ← تقدمت بمقدار 150°

$\boxed{V_{bc} = \sqrt{3} V_{bn} \sin(\omega t + 30°)}$ ← V_{bc} تقدمت بمقدار 30°

$\boxed{V_{ca} = \sqrt{3} V_{bn} \sin(\omega t - 90)}$ ← تأخرت بمقدار -90°

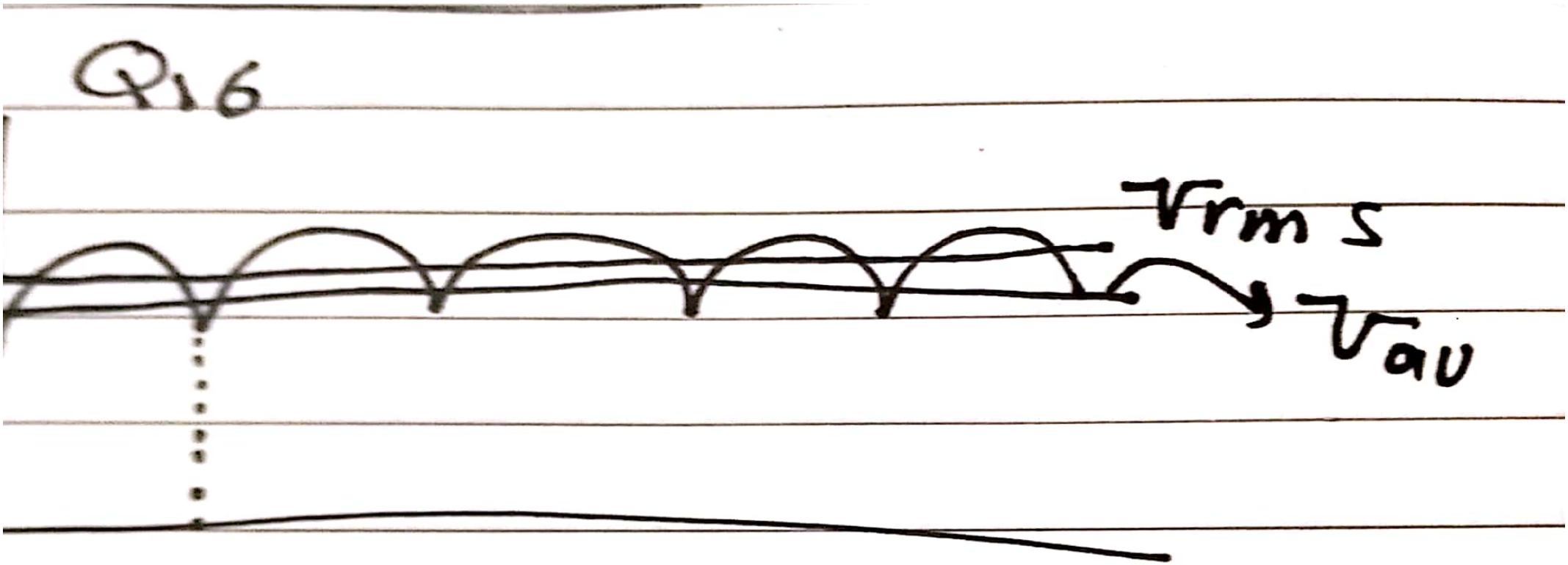
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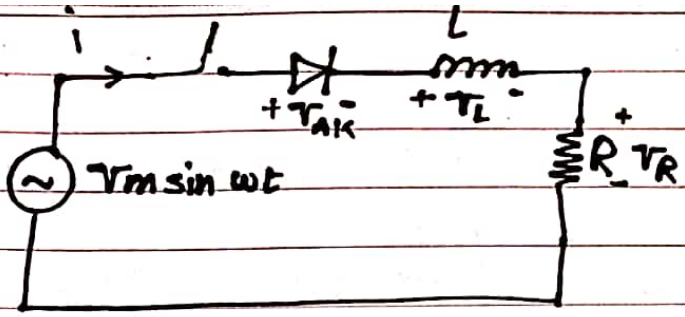


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Q16

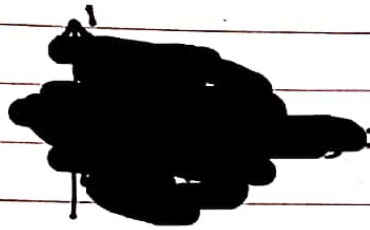


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عند اللحظة X أو اللحظة B تكون قيمة التيار = صفر

عند بداية التشغيل $i=0$ ← عند B (نهاية التشغيل) $i=0$



at $\omega t = X \rightarrow$ sw is on

$$-V_m \sin \omega t + V_L + V_R = 0$$

$$V_m \sin \omega t = V_L + V_R$$

$$V_m \sin \omega t = L\frac{di}{dt} + iR$$

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